

AI Tool Prompt Engineering Report

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AI Tool Overview

Name of AI tool: **GitHub Copilot**

Domain: **Software Engineering and Coding** with Ecosystem Integration

Reason for choice:

The task was to extend an existing **frontend website** project for Applications Development, another of my courses. Thus, a Coding AI was needed.

GitHub Copilot was convenient, integrated into my existing workflow and is available for use with a free tier. It is agentic, capable of analyzing the existing repository and creating new features in a similar manner as existing ones.

It also supports asking questions about the code, and planning out architectures, though I did not make use of these other features.

Initial Prompt

The setup begins at this commit of an existing public Git repository:

- <https://github.com/raffimolero/basicnextapp/tree/73f7388ee3e1c9b9e754932ce1594d762c2bac3b> - “add medical related tables and data”

The specifics of the setup are described under p14; Appendix - “Setup”. From there, I directed GitHub Copilot to use agent mode with the following files as context:

- postgresql/004-medical.sql
- app/dashboard/admin/users/page.tsx

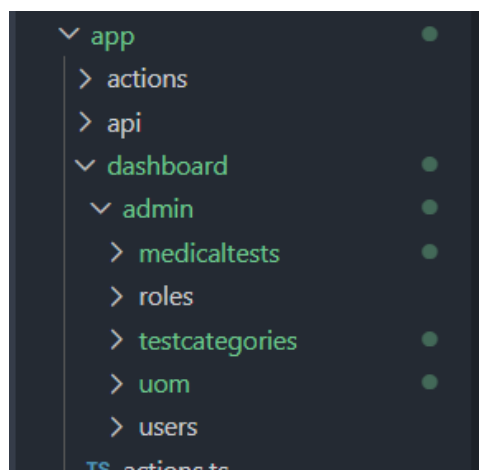
Create a folder called app/medical/

Given the UI style and auth guards under app/dashboard/admin/users/page.tsx, and following the schema in postgresql/004-medical.sql, write simple CRUD endpoints under the following:

- medical/uom/page.tsx
- medical/testcategories/page.tsx
- medical/medicaltests/page.tsx.

Note: under medicaltests/page.tsx, do not show `iduum` or `idcategory` numbers. Use SQL Joins to show the UOM Name and Category Name for all CRUD operations. When editing and creating rows, use dropdowns.

I then waited about 1 or 2 minutes for output. After some minor fixing (see p15; Appendix - “Initial Unrefined Prompt Output”), the file structure became as follows:

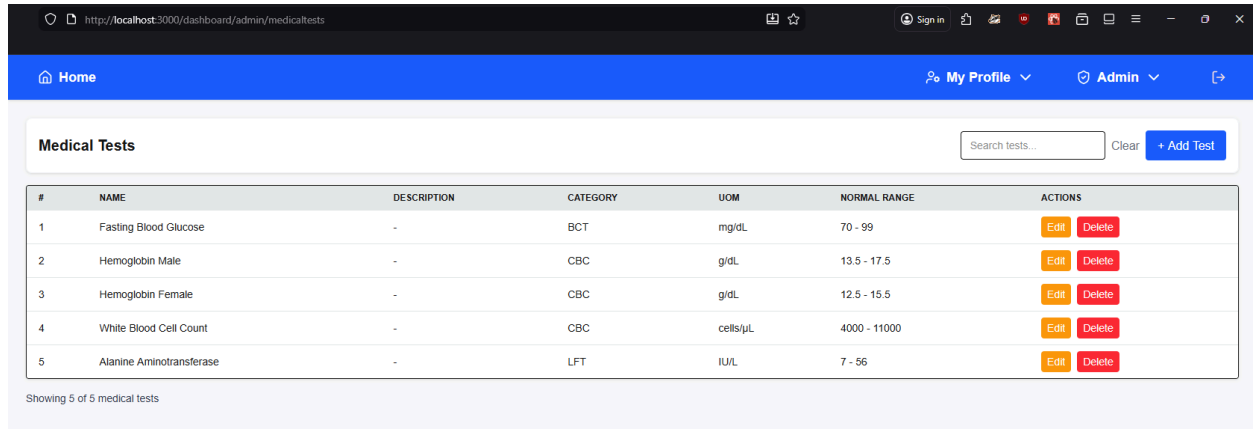


Unrefined Prompt Output

The result was a half-decent imitation of the existing **Roles** management dashboard, which was surprising, considering I specifically instructed it to copy the **Users** management page (see p17; Appendix - “Reference Pages”)

The design appeared fine, though it was missing hover transitions, and the interface for creating a new row was very different from the existing dashboard pages.

Copilot page (medicaltests):



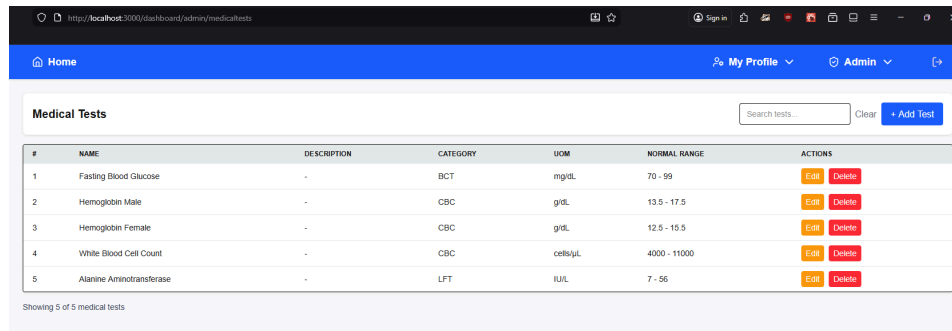
#	NAME	DESCRIPTION	CATEGORY	UOM	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose	-	BCT	mg/dL	70 - 99	Edit Delete
2	Hemoglobin Male	-	CBC	g/dL	13.5 - 17.5	Edit Delete
3	Hemoglobin Female	-	CBC	g/dL	12.5 - 15.5	Edit Delete
4	White Blood Cell Count	-	CBC	cells/ μ L	4000 - 11000	Edit Delete
5	Alanine Aminotransferase	-	LFT	IU/L	7 - 56	Edit Delete

Showing 5 of 5 medical tests

The next page shows the flow for creating a new Medical Test from the Copilot-generated page.

‘Add’ Flow - Medical Test (Copilot, Unrefined)

*Compare: p18; Appendix - “Add Flow: Roles (Original)”



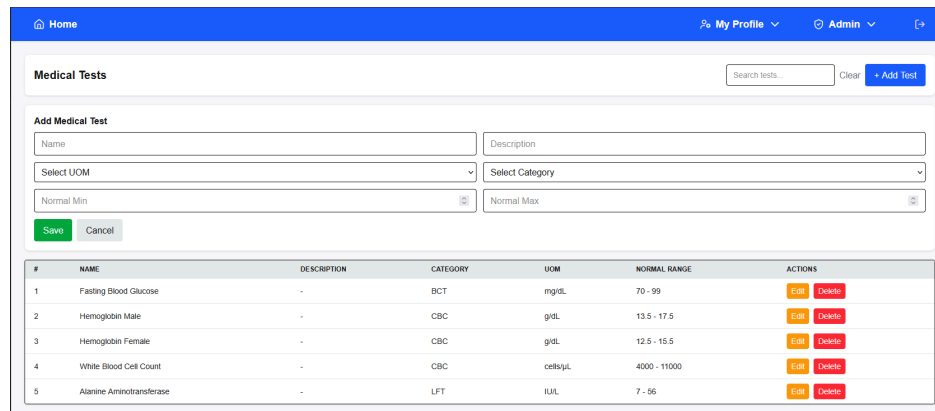
Medical Tests

Search tests... Clear + Add Test

#	NAME	DESCRIPTION	CATEGORY	UOM	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose	-	BCT	mg/dL	70 - 99	Edit Delete
2	Hemoglobin Male	-	CBC	g/dL	13.5 - 17.5	Edit Delete
3	Hemoglobin Female	-	CBC	g/dL	12.5 - 15.5	Edit Delete
4	White Blood Cell Count	-	CBC	cells/ μ L	4000 - 11000	Edit Delete
5	Alanine Aminotransferase	-	LFT	IU/L	7 - 56	Edit Delete

Showing 5 of 5 medical tests

After clicking “+ Add Test”:



Medical Tests

Search tests... Clear + Add Test

Add Medical Test

Name Description

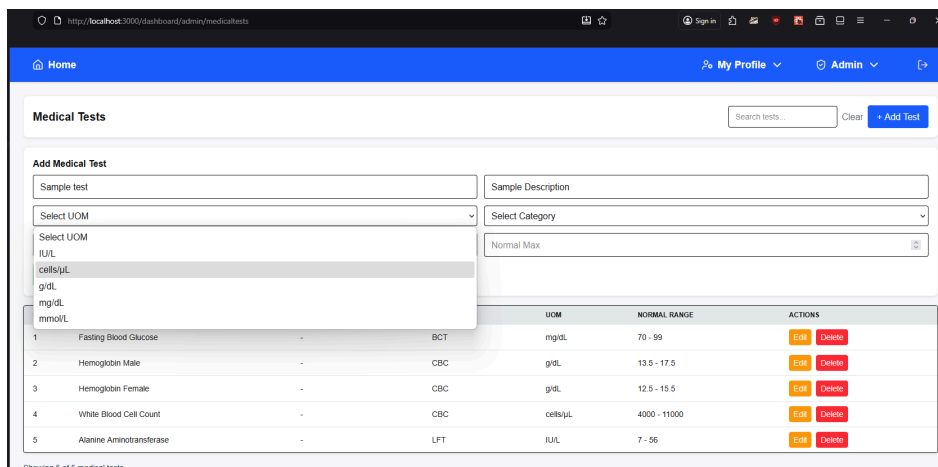
Select UOM Select Category

Normal Min Normal Max

[Save](#) [Cancel](#)

#	NAME	DESCRIPTION	CATEGORY	UOM	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose	-	BCT	mg/dL	70 - 99	Edit Delete
2	Hemoglobin Male	-	CBC	g/dL	13.5 - 17.5	Edit Delete
3	Hemoglobin Female	-	CBC	g/dL	12.5 - 15.5	Edit Delete
4	White Blood Cell Count	-	CBC	cells/ μ L	4000 - 11000	Edit Delete
5	Alanine Aminotransferase	-	LFT	IU/L	7 - 56	Edit Delete

I had expected the UI to blur and show a popup dialog, but instead a few text boxes, dropdowns, and a “Save” and “Cancel” button appeared. I filled in the details.



Medical Tests

Search tests... Clear + Add Test

Add Medical Test

Sample test Sample Description

Select UOM Select Category

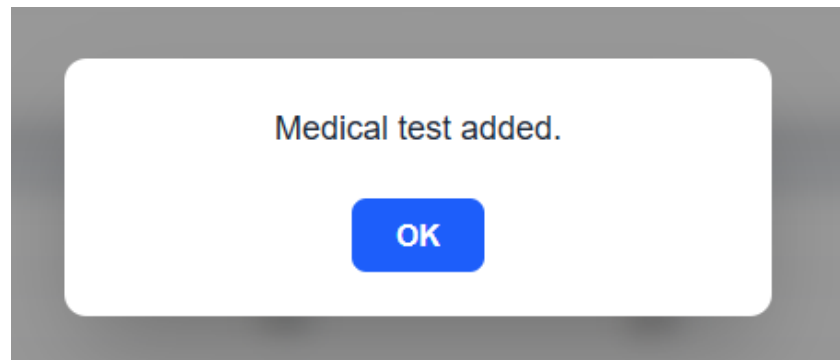
Normal Max

[Save](#) [Cancel](#)

#	NAME	DESCRIPTION	CATEGORY	UOM	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose	-	BCT	mg/dL	70 - 99	Edit Delete
2	Hemoglobin Male	-	CBC	g/dL	13.5 - 17.5	Edit Delete
3	Hemoglobin Female	-	CBC	g/dL	12.5 - 15.5	Edit Delete
4	White Blood Cell Count	-	CBC	cells/ μ L	4000 - 11000	Edit Delete
5	Alanine Aminotransferase	-	LFT	IU/L	7 - 56	Edit Delete

Showing 5 of 5 medical tests

After clicking “Save”:



The screen blurred and showed a notification popup, as expected.

After clicking “OK”:

#	NAME	DESCRIPTION	CATEGORY	UOM	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose	-	BCT	mg/dL	70 - 99	Edit Delete
2	Hemoglobin Male	-	CBC	g/dL	13.5 - 17.5	Edit Delete
3	Hemoglobin Female	-	CBC	g/dL	12.5 - 15.5	Edit Delete
4	White Blood Cell Count	-	CBC	cells/µL	4000 - 11000	Edit Delete
5	Alanine Aminotransferase	-	LFT	IU/L	7 - 56	Edit Delete
6	Sample test	Sample Description	CBC	cells/µL	50 - 100	Edit Delete

Showing 6 of 6 medical tests

The new medical test appears as expected.

For brevity, I will only describe the UX flows for “edit” and “delete”, as they are similar to the “add” flow: editing any of the previous rows simply pops up the text boxes and dropdowns again but filled in with the existing data. Saving the edited data shows a similar popup to saving new data. Deleting an item does not ask for confirmation, instead deleting the row immediately and only then showing a notification. The other 2 pages, “uom” and “testcategories”, also work the same way. Further screenshots would likely be unnecessary for them.

Summary

- The AI created working output.
- The AI copied a different page than specified.
- The outputs were visually similar, but lacked hover animations.
- The workflow was different, with text boxes appearing on the page itself instead of in a popup.
- The “Delete” button did not ask for additional confirmation.
- The code itself appeared similar to the existing code.

* That is, to say, both were vibe coded

* I don't think a single soul has actually read through the entire codebase

I concluded the initial prompt experiment by pushing a new GitHub branch. The relevant commit on that branch is here:

- <https://github.com/raffimolero/basicnextapp/tree/540de0a6b30ec6af14bb4b702003e27aa5fbae87>

Prompt Refinement

With the repository in that state, I turned to **ChatGPT** to refine my prompts further, given my hindsight. I first gave it context about this experiment, then instructed it to incrementally refine the prompt and keep the latest version and annotations in recent context at all times so that the prompt and further steps would always remain within its context window.

Full ChatGPT log here:

- <https://chatgpt.com/share/69c41ad0-2440-83a0-a778-e039c3d81622>
 - Note that I attempted to test ChatGPT's debugging capabilities by asking it to debug Copilot's code, though I had already fixed it by then and accidentally sent the fixed code. I cancelled its output and revealed the issue.

Full refined Copilot prompt here:

Create the following pages under `app/dashboard/admin/`:

- `medicaltests/page.tsx`
- `testcategories/page.tsx`
- `uom/page.tsx`

For each of the above folders, also create an `actions.ts` file.

Use `app/dashboard/admin/roles/page.tsx` and `app/dashboard/admin/roles/actions.ts` as the single source of truth.

Replicate the structure, layout, file organization, styling, and implementation patterns as closely as possible. Do not only imitate the visual design--match the component structure, layout hierarchy, Tailwind classes, and user interaction workflows.

When in doubt, copy and adapt code from `roles/page.tsx` instead of inventing new implementations.

File Structure Requirements:

- Each feature (`medicaltests`, `testcategories`, `uom`) must have:

- page.tsx
- actions.ts
- actions.ts must contain database helper functions (e.g., getX, createX, updateX, deleteX), following the same pattern as roles/actions.ts.
- page.tsx must:
 - import functions from its corresponding actions.ts
 - define fetchX using useCallback
 - define handlers like handleDeleteX inside the Page() component

General Requirements:

- Preserve the same layout structure, including placement of search bars, action buttons, and tables.
- Include search functionality if it exists in roles/page.tsx.
- Follow the same interaction patterns:
 - Adding a row
 - Editing a row
 - Deleting a row
- Do not introduce new UI patterns. Reuse only the patterns used in roles/page.tsx.
- Exclude features that are clearly specific to roles and not applicable to the target tables, but maintain the overall layout and spacing.

Styling (STRICT):

- Use Tailwind CSS.
- Reuse the same classNames and styling patterns from roles/page.tsx wherever possible.
- Match hover effects, spacing, alignment, and transitions.
- Do not invent new styling patterns unless absolutely necessary.

Buttons and Permissions:

- All action buttons (Add, Edit, Delete) must be wrapped using ButtonGuardWrapper.
- Follow the same usage pattern as in roles/page.tsx.
- For all newly created pages, use only:
 - requiredRoles={"ADMINISTRATOR"}

Modal Behavior (STRICT):

- All Create, Edit, and Delete actions must use modal dialogs. Do NOT render forms inline in the page.
- Modal behavior must match the existing Roles page.
- Duplicate the modal implementation pattern used in:
 - AddRoleModal.tsx
 - EditRoleModal.tsx

- DeleteRoleModal.tsx
- Create equivalent modal components for each new page (do not refactor or reuse existing ones).
- Each page should have its own Add/Edit/Delete modal components, even if code is duplicated.

Modal Requirements:

- Modals must:
 - Appear centered on screen
 - Render above a darkened and blurred background overlay
 - Match the same styling and structure as existing modals
- Add/Edit modals:
 - Prefill values when editing
 - Reset fields when creating
- Delete modals:
 - Require explicit user confirmation before deletion
 - Follow the same interaction flow as DeleteRoleModal.tsx

Data Fetching and Mutation (STRICT):

- Do NOT use API routes or server actions.
- Use direct database access via the existing query(...) function, as done in roles/actions.ts.
- Implement helper functions similar to:
 - getRoles
 - createRole
 - updateRole
 - deleteRole
- Fetch data on the client using useCallback and useEffect patterns similar to roles/page.tsx.
- Mutation pattern must follow:
 - Call async DB function (e.g., deleteX)
 - Show success/failure message using existing message system (e.g., showMessage)
 - Refetch data after mutation (no optimistic updates unless already used in roles/page.tsx)

Delete Behavior:

- After deletion, refetch and update the displayed data.

Database:

- Follow the schema defined in postgresql/004-medical.sql.

Foreign Key Handling:

- For fields like iduom and idcategory:
 - Do not display raw IDs in tables
 - Use SQL joins to display their corresponding names
 - In Create/Edit modals, use dropdown inputs populated from related tables

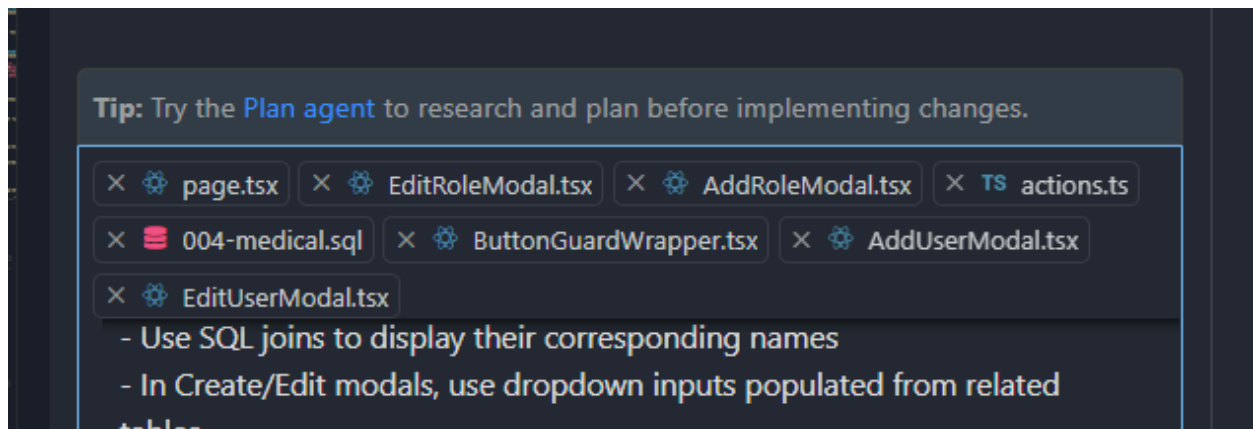
Navigation Integration:

- Add links to each new page under the admin dropdown menu in the main navigation.
- Follow the same structure and styling used for existing admin links.

Output Requirements:

- Generate complete working CRUD functionality.
- Do not include placeholders, TODOs, or incomplete logic.

Copilot Context files:



Refined Prompt Output

ROW #	NAME	DESCRIPTION	UOM	CATEGORY	NORMAL RANGE	ACTIONS
1	Alanine Aminotransferase		IU/L	LFT	7 - 56	Edit Delete
2	Fasting Blood Glucose		mg/dL	BCT	70 - 99	Edit Delete
3	Hemoglobin Female		g/dL	CBC	12.5 - 15.5	Edit Delete
4	Hemoglobin Male		g/dL	CBC	13.5 - 17.5	Edit Delete
5	White Blood Cell Count		cells/μL	CBC	4000 - 11000	Edit Delete

Showing 5 of 5 medical tests

After one prompt, it was almost exactly what I wanted. There was however one bug that affected both the UOM and Category columns, which I fixed soon after. (see p19; Appendix - “Refined Prompt: Copilot Quotation Error”)

The UX functioned mostly as expected, and the CSS styles worked, including any hover transitions. The same went for the “UOM” and “Test Categories” pages. Impressive for a one-shot prompt.

I then concluded the ChatGPT conversation and pushed the code to GitHub. The relevant commit can be found here:

- <https://github.com/raffimolero/basicnextapp/tree/3f8be387ece0f6c9a61b65833c37d5c344c6b1ff>

To keep the main page short, I will only include screenshots of the flow for adding a new Medical Test.

More screenshots can be found under:

- p20; Appendix - “Initial Refined Prompt Output”
- p21; Appendix - “Edit and Delete Flows: Roles (Original)”
- p22; Appendix - “Edit and Delete Flows: Medical Tests (Copilot)”

'Add' Flow: Medical Test (Copilot)

Add New Medical Test

Name

e.g. Fasting Blood Glucose

Description

Describe the medical test...

Unit of Measure

IU/L

Test Category

BCT

Normal Min

0

Normal Max

0

CAPSLOCK DETECTION ACTIVE

Add Medical Test

Cancel

Add New Medical Test

Name

Sample Test

Description

Sample Description

Unit of Measure

cells/ μ L

Test Category

CBC

Normal Min

50

Normal Max

100

CAPSLOCK DETECTION ACTIVE

Add Medical Test

Cancel

Medical test added successfully!

OK

Home

My Profile

Admin

Medical Test Management

Search medical tests...Clear

+ Add Medical Test

ROW #	NAME	DESCRIPTION	UOM	CATEGORY	NORMAL RANGE	ACTIONS
1	Alanine Aminotransferase		IU/L	LFT	7 - 56	EditDelete
2	Fasting Blood Glucose		mg/dL	BCT	70 - 99	EditDelete
3	Hemoglobin Female		g/dL	CBC	12.5 - 15.5	EditDelete
4	Hemoglobin Male		g/dL	CBC	13.5 - 17.5	EditDelete
5	Sample Test	Sample Description	IU/L	BCT	50 - 100	EditDelete
6	White Blood Cell Count		cells/ μ L	CBC	4000 - 11000	EditDelete

Showing 6 of 6 medical tests

Analysis

The difference was almost night-and-day. My initial prompt resulted in an interface that was visually similar to the wrong page and had different UX flows than desired. But one single refined prompt managed to get Copilot 99% of the way to my exact desired output, with the remaining 1% leaving me to debug a quotation mistake. Adding more context files helped narrow my requirements too.

The process of refining a prompt was also quite smooth. It felt like a “builder pattern” pipeline where I searched and sent information for the AI to organize for me. Then it would patch most ambiguities out. Granted, I did actually have to look around the codebase to find this information, but having the checklist laid out for me on a silver platter made the process feel much more efficient.

The new prompt is far more in-depth than I would write myself. It contains specific instructions organized in ways designed for an LLM/Agent to precisely follow. Every detail I had to give in the refinement process was a detail that I did not include in the original prompt, and there were many.

At this point this wasn’t just vibe coding anymore. I didn’t consider this seriously at first, but this is practically vibe prompting. Like, you just “vibe the vibe” and put up requirements for the AIs to churn through. Cool, but far from perfect. I still had to fix some mistakes, whether mine or from the AI. More importantly, with any AI use, it’s harder to learn in a meaningful way, and harder to understand the codebase, unlike when writing it from the ground up.

From here, I will keep these results in mind: proper prompt engineering can lead to far more accurate results than simply writing prompts by hand. Three hundredth word.

Appendix

Setup

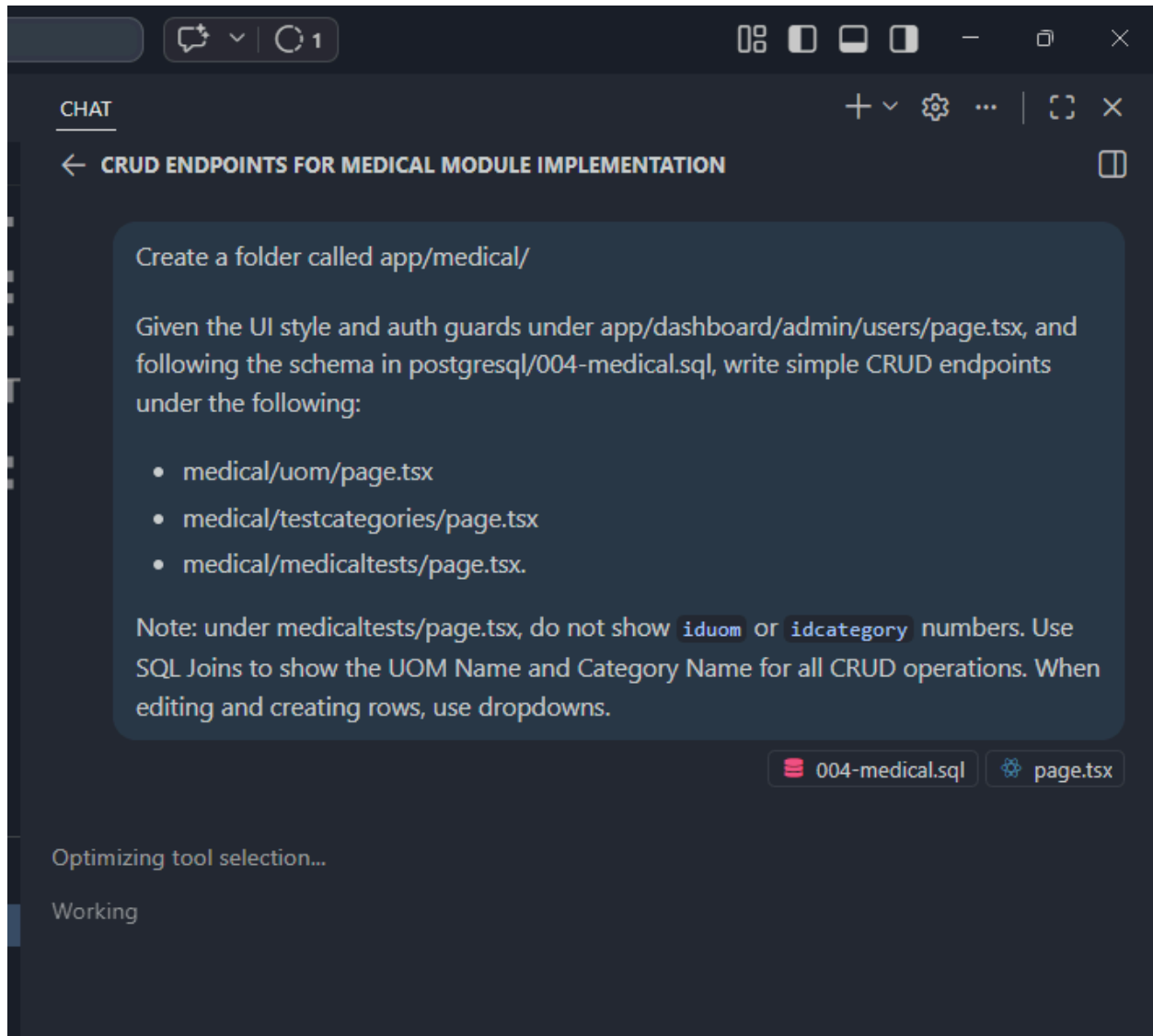
The public repository does not contain certain files containing sensitive data, such as any real “.env” files. However, the real “.env.local” file follows the template “myenvlocal.txt”, which is publicly available in the repository.

The project is assumed to have already been set up and runnable on a local machine, with npm packages already installed and an active local database, which should also have been set up with the SQL files having been run in the following order:

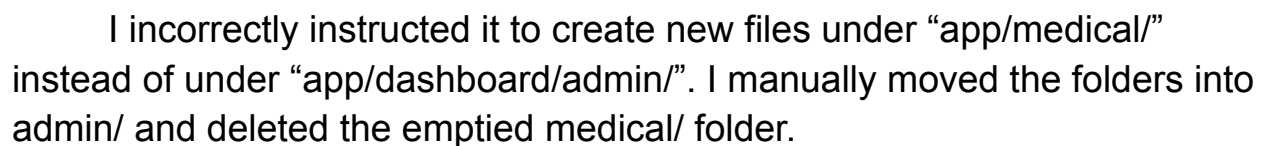
1. basicnextapp/postgresql/createtablesforbetterauth.sql
2. basicnextapp/postgresql/seedroles.sql
3. basicnextapp/postgresql/seedadminuser.sql
4. basicnextapp/postgresql/004-medical.sql

The default admin account is seeded with email “a@a.com” and password “password123”. It is not intended to be secure. Some user accounts and roles were also created through the interface, but they are not relevant for this assessment.

Initial Unrefined Prompt

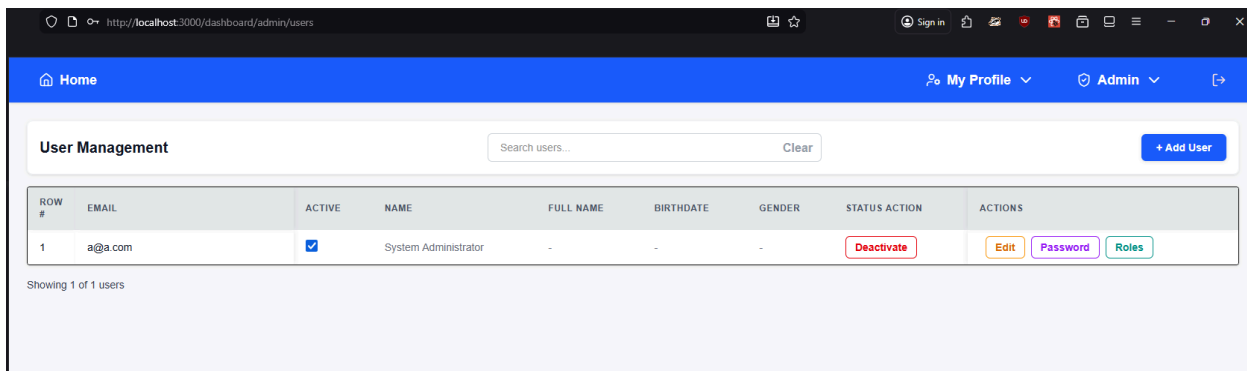


Importantly, I believe I misspoke in my initial prompt - I did not want “endpoints” (used only in backends) but rather “dashboards” within the frontend. GitHub Copilot appeared to understand what I actually wanted, though.



Reference Pages

Pre-existing users management page:

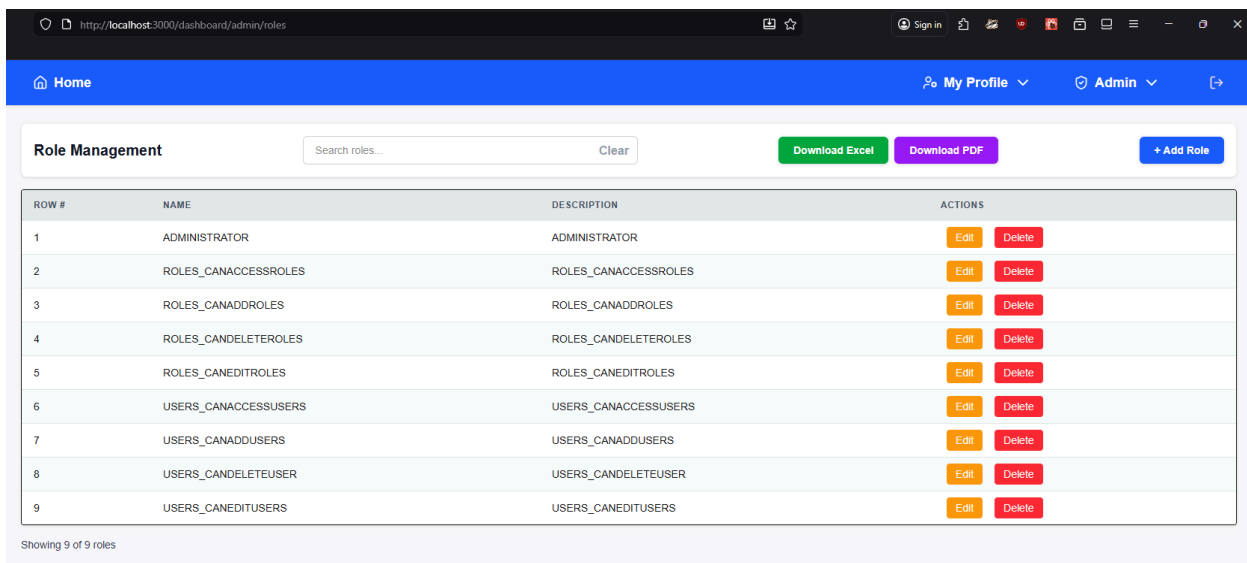


The screenshot shows the 'User Management' page in a web application. The page has a blue header with 'Home', 'My Profile', and 'Admin' links. Below the header, there's a search bar for users and a '+ Add User' button. The main content area contains a table with user information.

ROW #	EMAIL	ACTIVE	NAME	FULL NAME	BIRTHDATE	GENDER	STATUS ACTION	ACTIONS
1	a@a.com	☒	System Administrator	-	-	-	Deactivate	Edit Password Roles

Showing 1 of 1 users

Pre-existing roles management page:

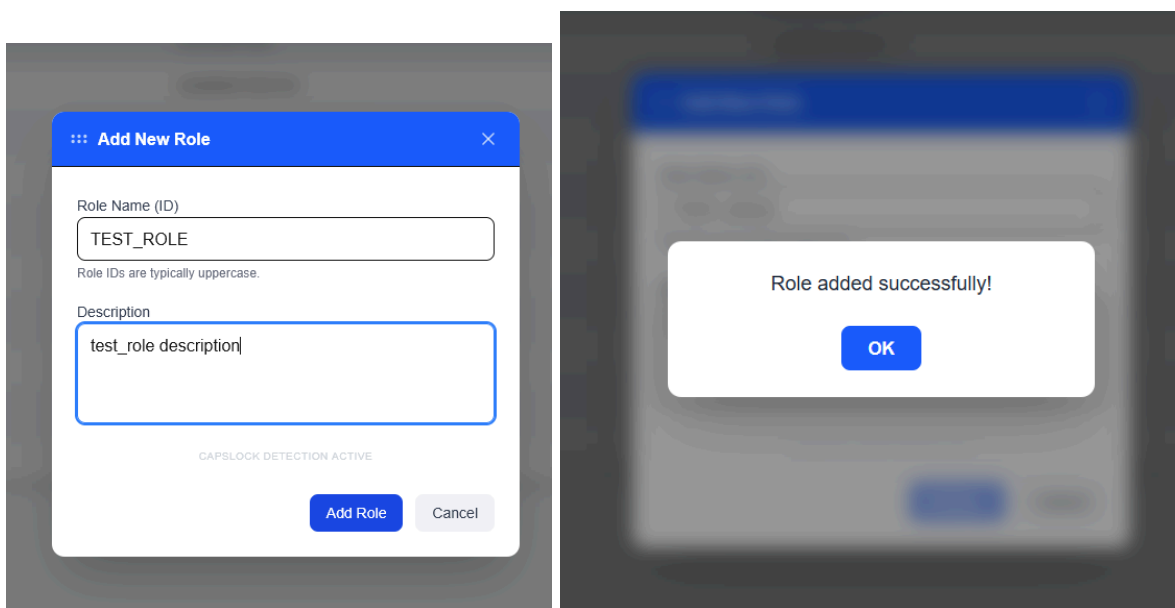


The screenshot shows the 'Role Management' page in a web application. The page has a blue header with 'Home', 'My Profile', and 'Admin' links. Below the header, there's a search bar for roles, 'Download Excel', 'Download PDF', and '+ Add Role' buttons. The main content area contains a table with role information.

ROW #	NAME	DESCRIPTION	ACTIONS
1	ADMINISTRATOR	ADMINISTRATOR	Edit Delete
2	ROLES_CANACCESSROLES	ROLES_CANACCESSROLES	Edit Delete
3	ROLES_CANADDROLES	ROLES_CANADDROLES	Edit Delete
4	ROLES_CANDELETEROLES	ROLES_CANDELETEROLES	Edit Delete
5	ROLES_CANEDITROLES	ROLES_CANEDITROLES	Edit Delete
6	USERS_CANACCESSUSERS	USERS_CANACCESSUSERS	Edit Delete
7	USERS_CANADDUSERS	USERS_CANADDUSERS	Edit Delete
8	USERS_CANDELETEUSER	USERS_CANDELETEUSER	Edit Delete
9	USERS_CANEDITUSERS	USERS_CANEDITUSERS	Edit Delete

Showing 9 of 9 roles

'Add' Flow: Roles (Original)



The screenshot shows the 'Role Management' dashboard. It includes a search bar, 'Download Excel' and 'Download PDF' buttons, and an '+ Add Role' button. The table below lists 10 roles, with the 6th role, 'TEST_ROLE', highlighted by a red box.

ROW #	NAME	DESCRIPTION	ACTIONS
1	ADMINISTRATOR	ADMINISTRATOR	Edit Delete
2	ROLES_CANACCESSROLES	ROLES_CANACCESSROLES	Edit Delete
3	ROLES_CANADDEROLES	ROLES_CANADDEROLES	Edit Delete
4	ROLES_CANDELETEROLES	ROLES_CANDELETEROLES	Edit Delete
5	ROLES_CANEDITROLES	ROLES_CANEDITROLES	Edit Delete
6	TEST_ROLE	test_role description	Edit Delete
7	USERS_CANACCESSUSERS	USERS_CANACCESSUSERS	Edit Delete
8	USERS_CANADDERUSERS	USERS_CANADDERUSERS	Edit Delete
9	USERS_CANDELETERUSER	USERS_CANDELETERUSER	Edit Delete
10	USERS_CANEDITUSERS	USERS_CANEDITUSERS	Edit Delete

Showing 10 of 10 roles

Refined Prompt: Copilot Quotation Error

Unfortunately, Copilot did not initially manage to properly retrieve the names of the UOM and Category that each test belonged to, displaying blank data and breaking search functionality. This was due to PostgreSQL automatically setting column names to lowercase unless quoted:

```
export async function getMedicalTests(): Promise<MedicalTest[]> {
  const { rows } = await query<MedicalTest>(`
    SELECT
      mt.id,
      mt.name,
      mt.description,
      u.name as uomName, -- Should be quoted "uomName"
      tc.name as categoryName, -- Should be quoted "categoryName"
      mt.normalmin,
      mt.normalmax,
      mt.iduom,
      mt.idcategory
    FROM public.medicaltests mt
    LEFT JOIN public.uom u ON mt.iduom = u.id
    LEFT JOIN public.testcategories tc ON mt.idcategory = tc.id
    ORDER BY mt.name ASC
  `);
  return rows;
}
```

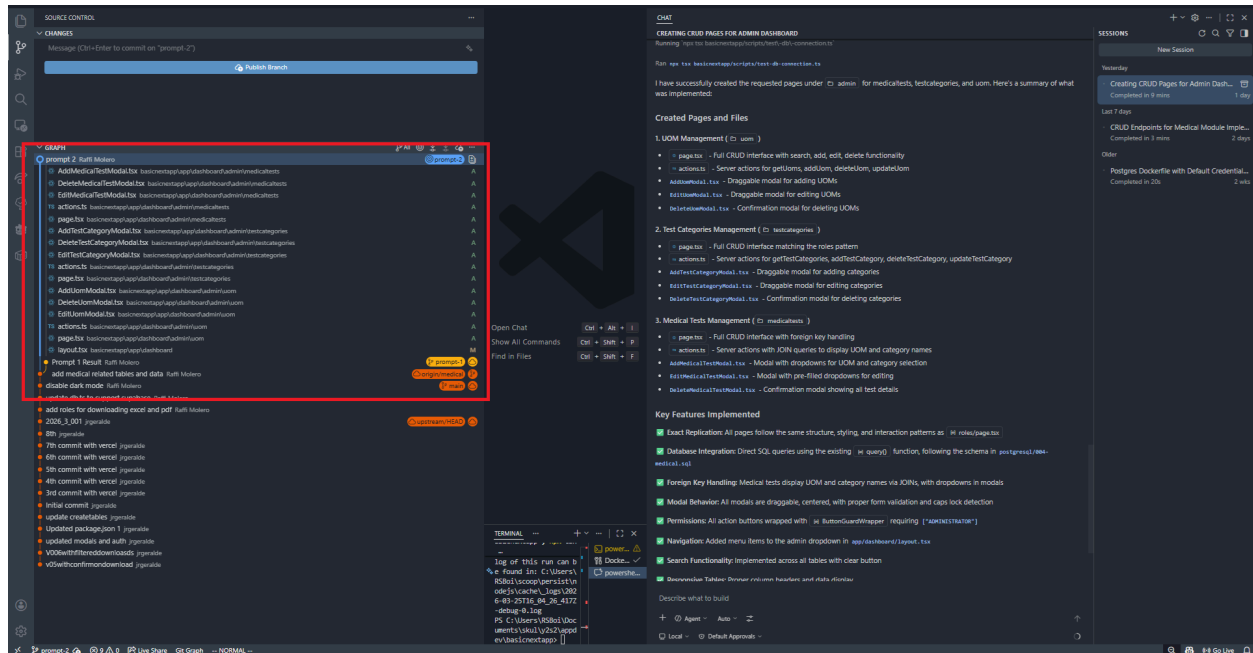
Neither Copilot nor the ChatGPT session caught this as a possible issue - Copilot likely because it was not given specific directives regarding PostgreSQL, and ChatGPT because it did not foresee this issue. I manually added the quotes and the UOM and Category columns started working properly.

Medical Test Management						
Search medical tests...			Clear		+ Add Medical Test	
ROW #	NAME	DESCRIPTION	UOM	CATEGORY	NORMAL RANGE	ACTIONS
1	Alanine Aminotransferase		IU/L	LFT	7 - 56	<button>Edit</button> <button>Delete</button>
2	Fasting Blood Glucose		mg/dL	BCT	70 - 99	<button>Edit</button> <button>Delete</button>
3	Hemoglobin Female		g/dL	CBC	12.5 - 15.5	<button>Edit</button> <button>Delete</button>
4	Hemoglobin Male		g/dL	CBC	13.5 - 17.5	<button>Edit</button> <button>Delete</button>
5	White Blood Cell Count		cells/ μ L	CBC	4000 - 11000	<button>Edit</button> <button>Delete</button>

Showing 5 of 5 medical tests

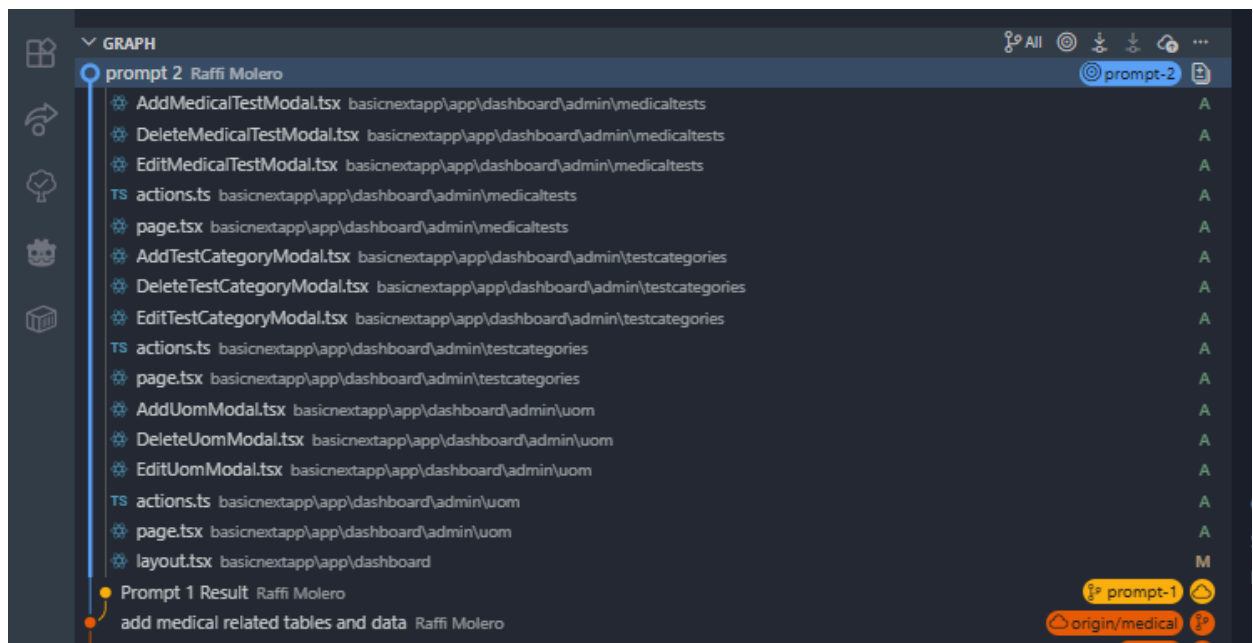
Initial Refined Prompt Output

Files added (see latest commit in git graph)



*pardon the username

Zoom for clarity



‘Edit’ and ‘Delete’ Flows: Roles (Original)

⋮ Edit Role

Role Name (ID)

SAMPLE

Role IDs cannot be changed.

Description

Sample Description|

CAPSLOCK DETECTION ACTIVE

Save Changes

Cancel

⋮ Delete Role

Role Name (ID)

SAMPLE

Description

Sample Description

Delete

Cancel

Role updated successfully!

OK

Are you sure you want to delete role "SAMPLE"?

Cancel

Yes, Delete

Role deleted successfully!

OK

Role Management

user| Clear

Download Excel

Download PDF

+ Add Role

ROW #	NAME	DESCRIPTION	ACTIONS	
1	USERS_CANACCESSUSERS	USERS_CANACCESSUSERS	Edit	Delete
2	USERS_CANADDUSERS	USERS_CANADDUSERS	Edit	Delete
3	USERS_CANDELETEUSER	USERS_CANDELETEUSER	Edit	Delete
4	USERS_CANEDITUSERS	USERS_CANEDITUSERS	Edit	Delete

Showing 4 of 9 roles

'Edit' and 'Delete' Flows: Medical Tests (Copilot)

Edit Medical Test

Name

Sample Test

Description

Sample Description

Unit of Measure

cells/ μ L

Test Category

CBC

Normal Min

50

Normal Max

100

CAPSLLOCK DETECTION ACTIVE

Save Changes

Cancel

Delete Medical Test

Name

Sample Test

Description

Sample Description

Unit of Measure

Test Category

Normal Min

50

Normal Max

100

Delete

Cancel

Medical test updated successfully!

OK

Are you sure you want to delete medical test "Sample Test"?

Cancel

Yes, Delete

Medical test deleted successfully!

OK

Medical Test Management						
			blood	Clear	+ Add Medical Test	
ROW #	NAME	DESCRIPTION	UOM	CATEGORY	NORMAL RANGE	ACTIONS
1	Fasting Blood Glucose		mg/dL	BCT	70 - 99	EditDelete
2	White Blood Cell Count		cells/ μ L	CBC	4000 - 11000	EditDelete

Showing 2 of 5 medical tests

The